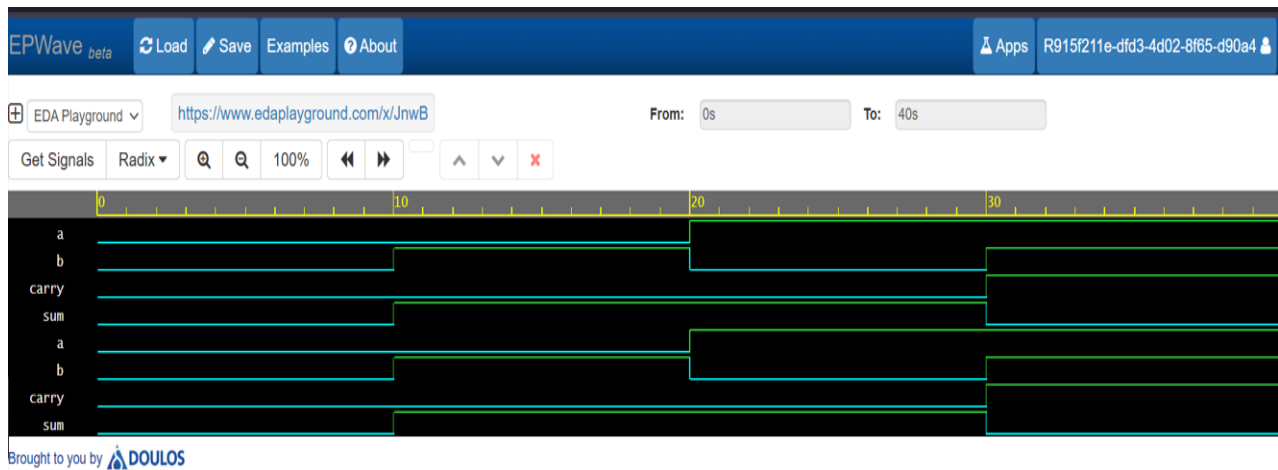
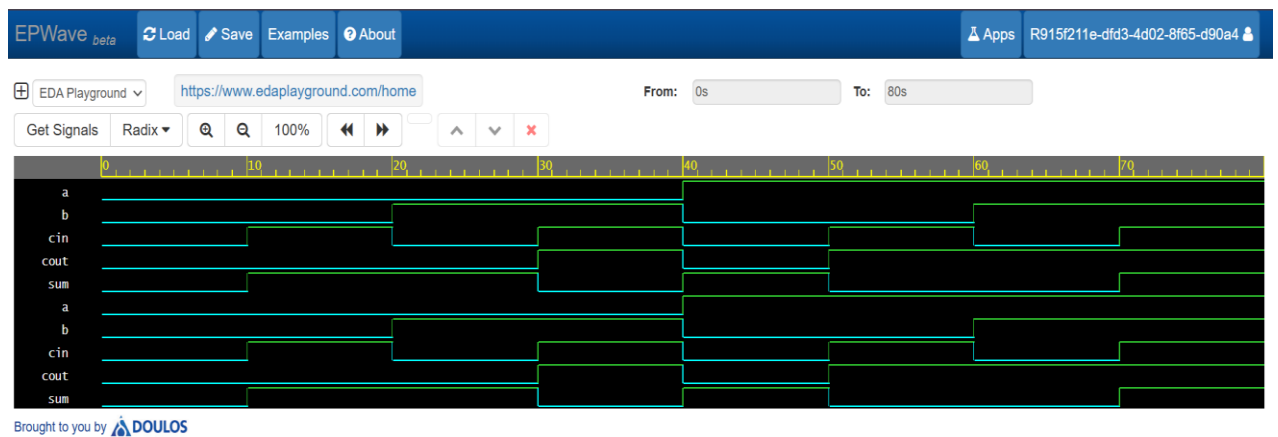


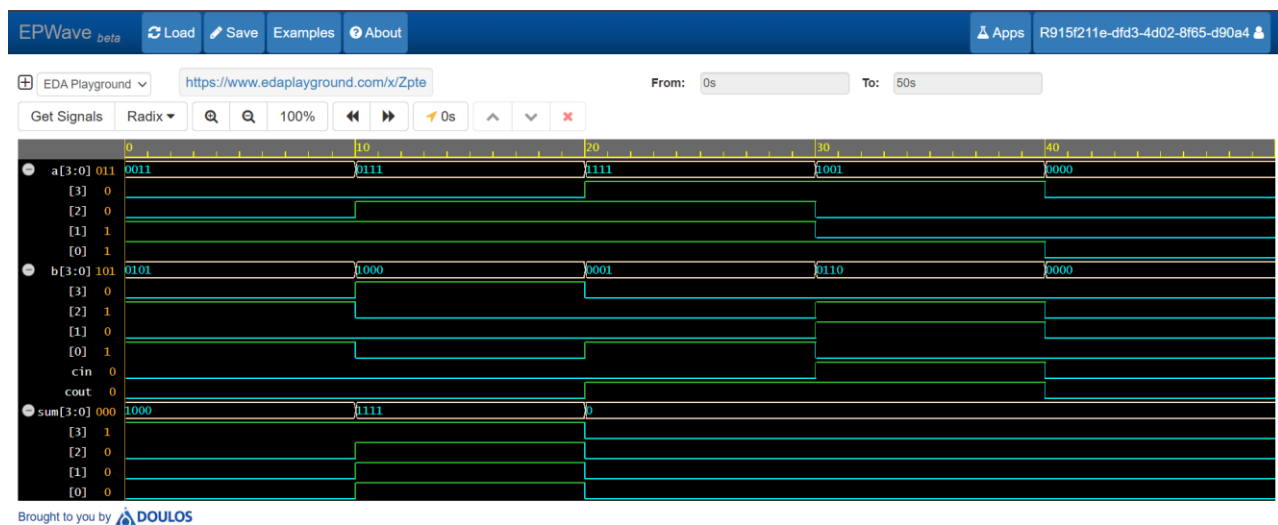
## 1. Half Adder :



## 2. Full Adder :



## 3. Ripple Carry Adder :



#### 4. 4-Bit ALU :

